

On Embedding Spaces

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What is an embedding?

In essence, it is some function of the form:

$$f_{\theta} : \mathcal{S} \rightarrow \mathbb{R}^n$$

where $\mathcal{S}$ is some input set, and $\mathbb{R}^n$ is some n -dimensional vector space, but where $|\mathcal{S}|$ is large and $n < |\mathcal{S}|$.

$$\begin{bmatrix} 0.1 \\ 0.3 \\ 1.2 \end{bmatrix} \in \mathbb{R}^3$$

Typically, it is also parametrized, as denoted by θ .

- Machine learning is typically a bunch of linear algebra, and thus it is often a prerequisite to represent your data as a vector.
- One could do one-hot encoding:

$$\{a, b, c\} \rightarrow a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

- But that doesn't work well if your input domain has many items.

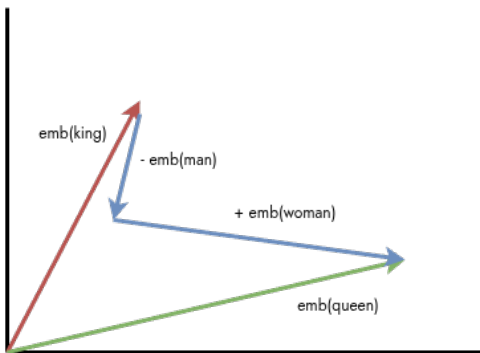
- Embeddings allow one to represent objects in a more compact number of dimensions.

$$\{a, b, c\} \rightarrow a = \begin{bmatrix} 0.7 \\ 2.1 \end{bmatrix}, b = \begin{bmatrix} 1.3 \\ 0.3 \end{bmatrix}, c = \begin{bmatrix} 0.01 \\ 5.6 \end{bmatrix}$$

- Embeddings have other interesting properties, namely that one can measure distance between them.

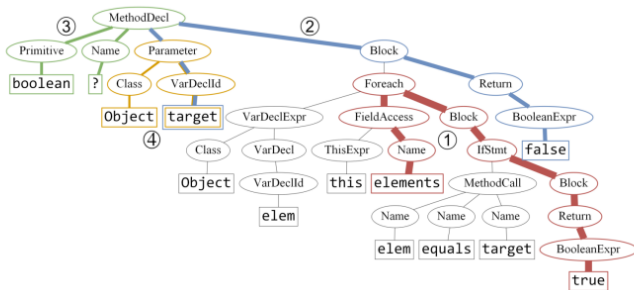
$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 \dots}$$

It has been shown that certain directions in these spaces can correspond to meaning!



- word2vec - Learn embeddings for words
- doc2vec - Learn embeddings for documents
- code2vec - Learn embeddings for code ASTs
- graph2vec - compute embeddings for graphs
- LLMs compute on embeddings of tokens!

Embedding ASTs

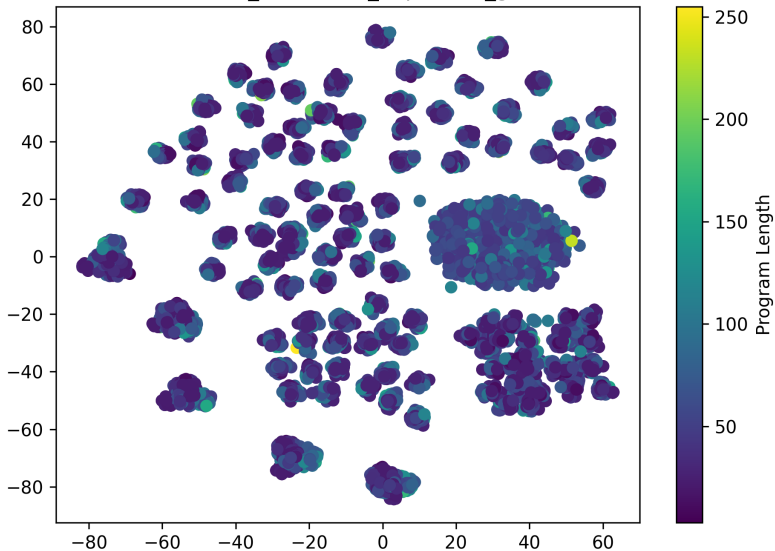


- Vector search (RAG)
- Clustering
- Downstream machine learning tasks (i.e., transformers/attention nowadays)

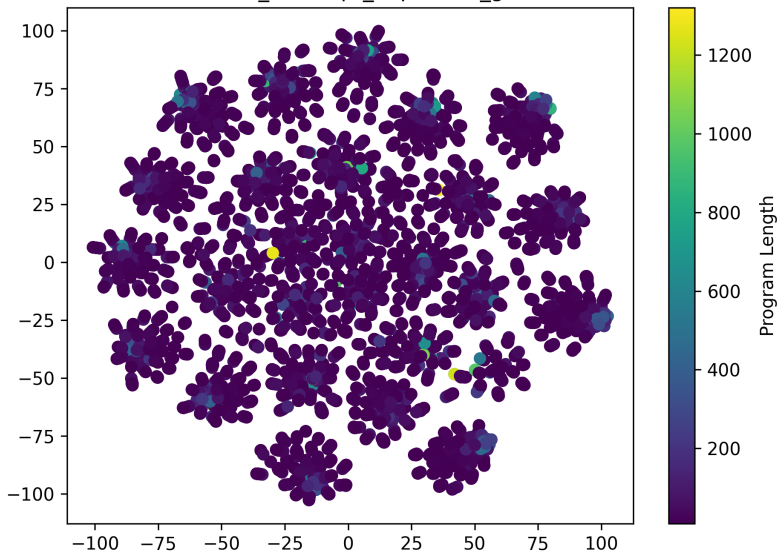
Inscrutability of latent spaces

The vectors contained in these spaces are weird!
Since they are very high-dimensional, it is hard to visualize how embeddings "group" based on meaning.
There are some tricks to try and get a look though...

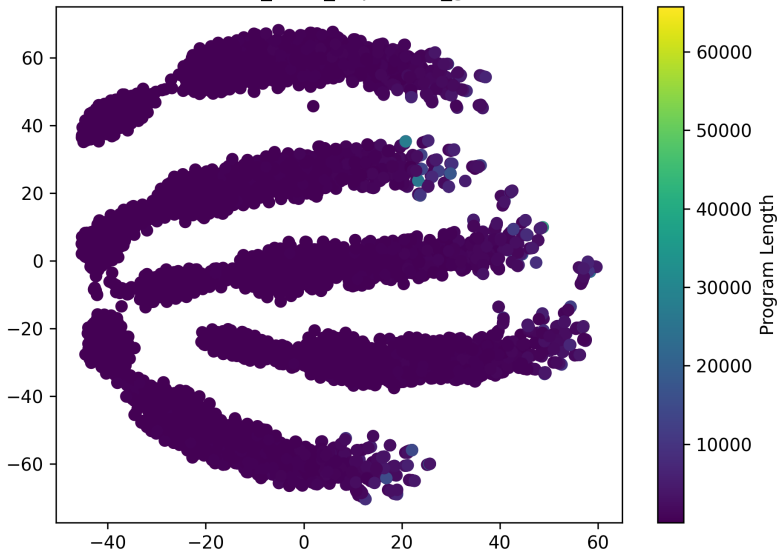
t-SNE (2D) of 10000_tacosched_nopartials_gensim dataset



t-SNE (2D) of 10000_tacoexpr_nopartials_gensim dataset



t-SNE (2D) of 5000_karel_nopartials_gensim dataset



- Mechanistic interpretability looks to find methods for understanding structures like latent spaces.